Median, Quartiles and the Five-Point Summary

Name _____ Date _____ No calculator Version 1

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KEY WORDS median *(tau waenga)* · lower quartile *(hauwhā raro)* · upper quartile *(hauwhā runga)*

A The median ●○○

Remember: Put the values in order first. The median is the middle value. With an even number of values, it is halfway between the middle two.

1. Find the median of each set of data.

a) 3, 8, 5

b) 9, 2, 7, 4, 6

c) 4, 10, 6, 8

d) 12, 5, 9, 3, 7, 10

e) 15, 11, 19, 14, 12, 17, 13

f) 2, 2, 5, 3, 8, 1

g) 20, 35, 15, 40, 25, 30, 10, 45

h) 6.5, 4, 7, 5.5, 8

i) 7, 1, 9, 3, 6, 8

j) 30, 10, 20

k) 18, 22, 18, 25, 30, 21

l) 7, 0, 3, 9, 4, 4, 6, 2, 8

m) 100, 85, 120, 95

n) −3, 4, 0, −6, 2

B Quartiles ●●○

Remember: The lower quartile (LQ) is the median of the lower half of the data. The upper quartile (UQ) is the median of the upper half. If there is a middle value, leave it out of both halves.

2. These sets of data are already in order. Find the lower quartile and the upper quartile.

a) 1, 3, 4, 6, 8, 9, 12

b) 2, 5, 6, 8, 10, 11, 13, 16

c) 4, 5, 7, 9, 10, 12, 15, 16, 20

d) 10, 11, 13, 14, 16, 19

e) 3, 4, 4, 6, 7, 9, 10, 12, 13, 15

f) 21, 25, 28, 30, 33, 36, 40, 42, 45, 47, 50

g) 5, 8, 9, 14, 15

h) 0, 2, 2, 3, 5, 6, 6, 8, 9, 9, 10, 12

C The five-point summary ●●●

Remember: The five-point summary is the minimum, lower quartile, median, upper quartile and maximum. Range = maximum − minimum. Interquartile range (IQR) = UQ − LQ.

3. The numbers of goals a netball team scored in its last nine games: 24, 31, 18, 27, 35, 22, 29, 40, 26

a) Find the five-point summary.

b) Find the range.

c) Find the interquartile range.

4. The ages, in years, of the eleven players in a football team: 14, 15, 13, 16, 14, 15, 17, 13, 14, 16, 15

a) Find the five-point summary.

b) Find the range.

c) Find the interquartile range.

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5. The times, in seconds, that eight students took to run 100 m: 15, 13, 17, 14, 16, 19, 14, 18

a) Find the five-point summary. **b)** Find the range. **c)** Find the interquartile range.

6. The numbers of books ten students read in a term: 3, 7, 1, 5, 12, 4, 6, 2, 8, 5

a) Find the five-point summary. **b)** Find the range. **c)** Find the interquartile range.

D Range and interquartile range ●●○

7. Find the range and the interquartile range for each five-point summary.

a) Minimum 4, LQ 9, median 13, UQ 18, maximum 25

b) Minimum 12, LQ 20, median 26, UQ 31, maximum 40

c) Minimum 0, LQ 3, median 5, UQ 9, maximum 14

d) Minimum 45, LQ 58, median 64, UQ 72, maximum 90

e) Minimum 1.5, LQ 2.5, median 3, UQ 4, maximum 6.5

f) Minimum 100, LQ 140, median 165, UQ 190, maximum 260

g) Minimum 8, LQ 8, median 9, UQ 11, maximum 15

h) Minimum 20, LQ 35, median 50, UQ 65, maximum 80

i) Minimum 3, LQ 7, median 12, UQ 16, maximum 22

★ Extension

8. Five numbers in order are 3, 5, x, 9, 10. Their median is 7. What is x?

9. Six numbers in order are 3, 5, 6, y, 11, 30. Their median is 7. Find y.

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Median, Quartiles and the Five-Point Summary

Name _____ Date _____ No calculator Version 2 Scan to practise this topic online

KEY WORDS median *(tau waenga)* · lower quartile *(hauwhā raro)* · upper quartile *(hauwhā runga)*

A The median ●○○

Remember: Put the values in order first. The median is the middle value. With an even number of values, it is halfway between the middle two.

1. Find the median of each set of data.

a) 4, 9, 6

b) 8, 3, 7, 5, 1

c) 5, 11, 7, 9

d) 13, 6, 10, 2, 8, 11

e) 16, 12, 20, 15, 11, 18, 14

f) 3, 3, 6, 4, 9, 2

g) 25, 40, 10, 45, 30, 35, 15, 50

h) 7.5, 5, 8, 6.5, 9

i) 8, 2, 10, 4, 7, 9

j) 40, 10, 30

k) 19, 23, 19, 26, 31, 22

l) 8, 0, 4, 9, 5, 5, 7, 3, 6

m) 110, 90, 130, 105

n) −4, 5, 0, −7, 3

B Quartiles ●●○

Remember: The lower quartile (LQ) is the median of the lower half of the data. The upper quartile (UQ) is the median of the upper half. If there is a middle value, leave it out of both halves.

2. These sets of data are already in order. Find the lower quartile and the upper quartile.

a) 2, 4, 5, 7, 9, 10, 14

b) 1, 3, 6, 7, 9, 12, 14, 15

c) 3, 6, 8, 9, 11, 13, 14, 17, 21

d) 12, 13, 15, 16, 18, 22

e) 2, 5, 5, 7, 8, 9, 11, 13, 14, 18

f) 20, 24, 27, 31, 32, 35, 39, 41, 44, 46, 52

g) 4, 7, 10, 13, 18

h) 0, 1, 3, 3, 4, 6, 7, 7, 8, 10, 11, 13

C The five-point summary ●●●

Remember: The five-point summary is the minimum, lower quartile, median, upper quartile and maximum. Range = maximum − minimum. Interquartile range (IQR) = UQ − LQ.

3. The numbers of goals a netball team scored in its last nine games: 26, 33, 19, 28, 37, 21, 30, 42, 25

a) Find the five-point summary.

b) Find the range.

c) Find the interquartile range.

4. The ages, in years, of the eleven players in a football team: 13, 15, 14, 16, 13, 15, 17, 14, 14, 16, 12

a) Find the five-point summary.

b) Find the range.

c) Find the interquartile range.

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5. The times, in seconds, that eight students took to run 100 m: 16, 14, 18, 13, 17, 20, 15, 19

a) Find the five-point summary.

b) Find the range.

c) Find the interquartile range.

6. The numbers of books ten students read in a term: 4, 8, 2, 6, 11, 3, 7, 1, 9, 6

a) Find the five-point summary.

b) Find the range.

c) Find the interquartile range.

D Range and interquartile range ●●○

7. Find the range and the interquartile range for each five-point summary.

a) Minimum 5, LQ 8, median 12, UQ 19, maximum 27

b) Minimum 15, LQ 22, median 28, UQ 34, maximum 42

c) Minimum 0, LQ 2, median 6, UQ 8, maximum 15

d) Minimum 40, LQ 55, median 63, UQ 70, maximum 88

e) Minimum 1.5, LQ 3, median 3.5, UQ 4.5, maximum 7

f) Minimum 120, LQ 150, median 175, UQ 210, maximum 280

g) Minimum 6, LQ 6, median 8, UQ 10, maximum 14

h) Minimum 25, LQ 40, median 55, UQ 60, maximum 85

i) Minimum 2, LQ 6, median 11, UQ 17, maximum 24

★ Extension

8. Five numbers in order are 4, 6, x, 11, 13. Their median is 8. What is x?

9. Six numbers in order are 4, 6, 7, y, 13, 33. Their median is 8. Find y.

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